

**INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
MID-SPRING SEMESTER EXAMINATION 2023-24**

Subject No. : CS60075

Subject : Natural Language Processing

Duration: 2 Hours

Full Marks: 60

Department: Computer Science and Engineering

Specific charts, graph paper, log book etc., required: NA

Special Instructions (if any) :

- *Attempt all questions. All parts of the same question must be answered together.*
 - *Answers should be brief and to-the-point. Sketchy answers will be penalized.*
 - *Make reasonable assumptions only if necessary, and state any assumptions made. No clarification can be provided in the examination hall.*
 - *All working steps must be shown. Writing the answer without showing steps will not be given any credit, even if the answer is correct.*
 - *Use base 10 for all log calculations. You can use calculators.*
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Question 1 (Miscellaneous) [8 x 2 = 16]

State whether the following statements are True or False, giving a short (1-2 sentences) meaningful explanation. Marks will not be awarded unless the explanation is correct.

- (a) Stemming mostly generates linguistically correct words.
- (b) Removing stop-words always improves the accuracy of NLP models.
- (c) A n -gram language model predicts the probability of a word based on the previous n words.
- (d) Smoothing techniques are used in n -gram models to handle unseen n -grams.
- (e) Higher-order n -gram models always perform better than lower-order models.
- (f) In HMM-based n -gram language models, the hidden states represent words.
- (g) Word2Vec uses one-hot encoding to represent words in its input and output vector spaces.
- (h) In a dependency graph, every word must depend on exactly one parent except for the root word.

Question 2 (Language Models) [2 + 3 + 2 + 6 + 2 = 15]

Consider the following pre-processed corpus C with all stopwords removed:

<s> Trump become President US </s>
<s> Obama Trump Biden earlier US President </s>
<s> Trump defeat Harris earlier Hilary </s>

Also consider the pre-processed sentence S :

<s> Biden defeat Trump become US President </s>

- (a) Calculate all the unigram probabilities. Remember that $\langle s \rangle$ and $\langle /s \rangle$ are also unigrams.
 (b) What is the perplexity of the unigram LM when evaluated over sentence S ?
 (c) Calculate all the bigram probabilities for all bigrams in the corpus. Remember that $\langle /s \rangle \langle s \rangle$ is not a valid bigram.
 (d) **Good-Turing smoothing:** Calculate the perplexity of S for the bigram LM after performing the following smoothing:

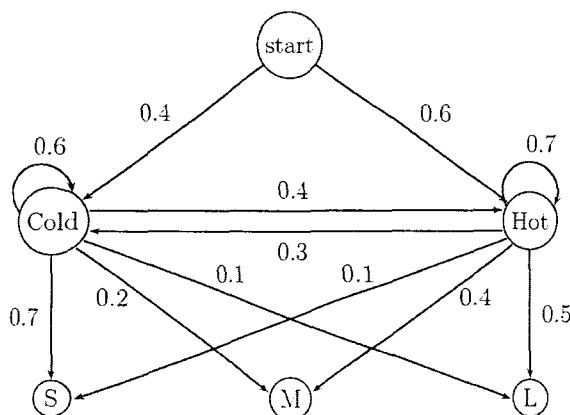
$$P_{bi-GT}(w_i | w_{i-1}) = \frac{(r+1) \cdot N(r+1)}{N(r) \cdot N}$$

given that $r = C(w_{i-1}w_i)$ is the number of times the bigram $w_{i-1}w_i$ occurs in the corpus, $N(r)$ is the total number of bigrams that occur r times in the corpus, and N is the total number of bigrams in the corpus. You can take $N(0) = 1$.

- (e) What is the biggest drawback of the Good-Turing Smoothing technique? Suggest one way to overcome this. [Hint: Think about what happens to $N(r)$ for larger values of r]

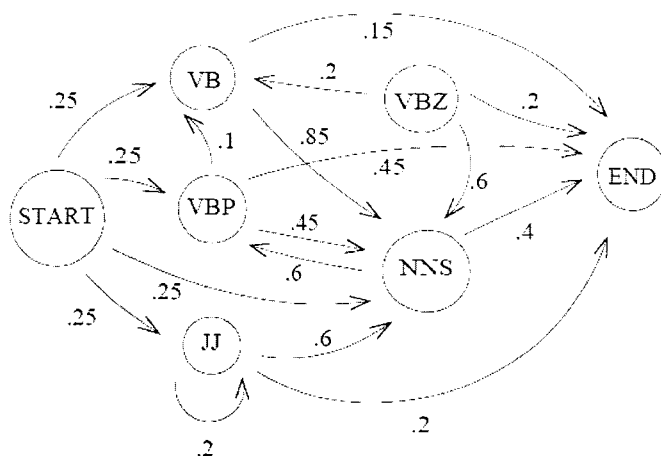
Question 3 (POS tagging and HMMs) [7 + 4 + (1 + 3) = 15]

- (a) Bob wants to determine the average annual temperature at a particular location by looking into the indirect evidence of the temperature in terms of the size of tree growth rings. Assume two temperatures Cold and Hot, and three different sizes of tree rings: small, medium, and large; denoted by S, M, L. Consider the HMM shown below, where Cold and Hot are the hidden states, and S, M, L are the observations/outputs.



Suppose that an observed sequence of tree growth rings is (S, M, S, L). What is the most likely sequence of temperatures over a four-year period which yields the observations (S, M, S, L)? Show the calculations in detail.

- (b) Calculate the probability that the sequence of words “*railway administrations*” will be assigned the parts of speech: *VB NNS*
 The transition probabilities (given in the figure) and emission/observation probabilities of words from the states (given in the table) are as follows:



	<i>railway</i>	<i>administrations</i>
JJ	0.158	0
NNS	0	0.0385
VB	0.145	0
VBP	0.081	0
VBZ	0	0.005

(c) Consider the following corpus:

their/N raises/N <stop>
 he/N raises/V <stop>
 my/N purses/N <stop>

Note that N and V are the POS tags of the words in those sentences.

- List the maximum-likelihood initial state probabilities for an HMM estimated from this data.
- List the maximum-likelihood transition probabilities for an HMM estimated from this data (including transitions to the STOP symbol).

Question 4 (Word embeddings) [(2+2+2) + (4+2) + 2 = 14]

(a) Consider the following paragraph:

The **band** played in the **garage**, which was their favourite place.
 The **garage** echoed with the **band** 's music and was filled with their equipment and cars.
 The **band** loved practicing and making music there, spending hours bonding.
 The **garage** was perfect for the **band**, and they found inspiration there.
 Sometimes, their cars were parked in the **garage**, as they played their favourite music.
 The **band** members loved chatting about their music, cleaning their equipment.
 The **garage** witnessed the **band** 's growth, serving as their sanctuary and holding many memories of their time together.

- Write down the term co-occurrence based word embedding for words **band** and **garage**, based on the vocabulary words: **music, cars, played, favourite, equipment, loved, practicing**. Consider each sentence as the context.
- Calculate the cosine similarity of the vectors, if $|X|$ is the length of the vector X (sq root of sum of squares) and $\text{cossim}(A, B) = A \cdot B / (|A| \cdot |B|)$
- Calculate the Pointwise Mutual Information (PMI) between these two words.

- (b) You are training a SkipGram word2vec model. Consider a vocabulary of 8 words in this order: **{all, eat, couch, bed, sleep, cats, dogs, peace}**. Each of these words has the same value of in and out weight vectors at a time step, which are as follows (in order): $[2 \ 2 \ -1]$, $[2 \ 0 \ 1]$, $[-2 \ 2 \ -2]$, $[1 \ -1 \ 2]$, $[-1 \ -1 \ 1]$, $[-1 \ 2 \ 0]$, $[2 \ -2 \ -1]$, $[0 \ 1 \ -2]$.

Suppose at this time step, the sliding window contains **[all cats sleep couch peace]**, where **sleep** is the centre word and others are context words. We are using a variation of SkipGram, where all context words are considered together as the label / ground truth for calculating the cross-entropy loss. Calculate the **(i)** values at the output before and after softmax, and **(ii)** cross-entropy loss at this time step.

- (c) Consider 3-dimensional word embeddings of the words **f1, football, hamilton, sachin, argentina** as $[3 \ 2 \ -1]$, $[3 \ 2 \ 1]$, $[-5 \ 6 \ -3]$, $[1 \ -4 \ 2]$, $[-2 \ 2 \ 3]$. Utilizing the analogy among word embeddings, find the closest embedding of the word **india**, given that messi is from argentina, while sachin is from india, and that messi plays football while hamilton races in f1.
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